

Applying Data Quality Dimensions

[Exercise]

In this exercise, you'll be given a dataset to which you'll apply each of the six dimensions of data quality to evaluate its integrity. You are required to document your findings and suggest improvements based on your assessment.

Learning Objectives

By the end of this train, you should be able to:

- Evaluate a dataset across the six dimensions of data quality and recommend improvements.

Exercise Details

Dataset ([link](#)): A sample dataset in Google Sheets format containing sales records, customer details, and sensor measurements. The dataset may have issues related to the six data quality dimensions.

Tools to use: Google Sheets (or another spreadsheet tool like Excel)

Deliverable: Document your findings and recommendations.

Instructions

1. **Copy the dataset:** Open and make a copy of the provided Google Sheet.
2. **Review the dataset:** Examine the dataset carefully for potential data quality issues.
3. **Apply the six dimensions of data quality:** For each dimension, assess the dataset and record your findings.

Dimensions to Evaluate

Accuracy:

- **Check:** Are the data points an accurate reflection of real-world events? Are there erroneous or impossible values (e.g., negative sales numbers, incorrect sensor readings)?
- **Questions to ask:** Are any data points inconsistent with the real-world context they represent? Are there obvious mistakes such as misformatted numbers or out-of-range values?
- **Findings:** List any inaccuracies and suggest corrective actions.

Consistency:

- **Check:** Are the data entries consistent across different columns or sources? Do similar data points appear with different formatting or units?
- **Questions to ask:** Are there discrepancies in how the data is represented (e.g., different units or inconsistent formatting)?
- **Findings:** Document any inconsistencies and propose ways to standardise the data.

Timeliness:

- **Check:** Is the data up-to-date? Are there any records that are outdated or no longer relevant?
- **Questions to ask:** Are there records with outdated timestamps or irrelevant data?
- **Findings:** Highlight any timeliness issues and suggest how to ensure data stays current.

Validity:

- **Check:** Does the data conform to predefined rules and formats? Are there values that violate expected formats (e.g., an email field without "@")?
- **Questions to ask:** Are all data entries valid (e.g., correct email format, appropriate values for numerical fields)?
- **Findings:** List any invalid data and suggest validation rules to improve the dataset.

Completeness:

- **Check:** Are there missing values in the dataset? Is any essential information incomplete?
- **Questions to ask:** Are there any fields or rows with missing data? Are there patterns of missing data in certain columns?
- **Findings:** Identify any missing data and suggest strategies for filling gaps (e.g., data imputation, requesting missing values).

NOTE: Distinguish between *essential* missing data (critical to the analysis objectives) and *non-essential* missing data. Examples:

- Essential: Customer IDs in sales records or timestamps in sensor measurements.
- Non-essential: Optional customer feedback or supplementary notes.

Uniqueness:

- **Check:** Are there any duplicate records? Does the dataset contain repeated entries for the same entity?
- **Questions to ask:** Are there duplicate rows or repeated values in the dataset?
- **Findings:** List any duplicates and suggest methods for deduplication.

Document Findings and Recommendations

For each dimension, summarise your findings and document any issues you encountered. Provide actionable recommendations including:

- Standardising data formats.

- Setting up automatic validation rules.
- Filling in missing values using imputation techniques.
- Removing or flagging duplicate entries.